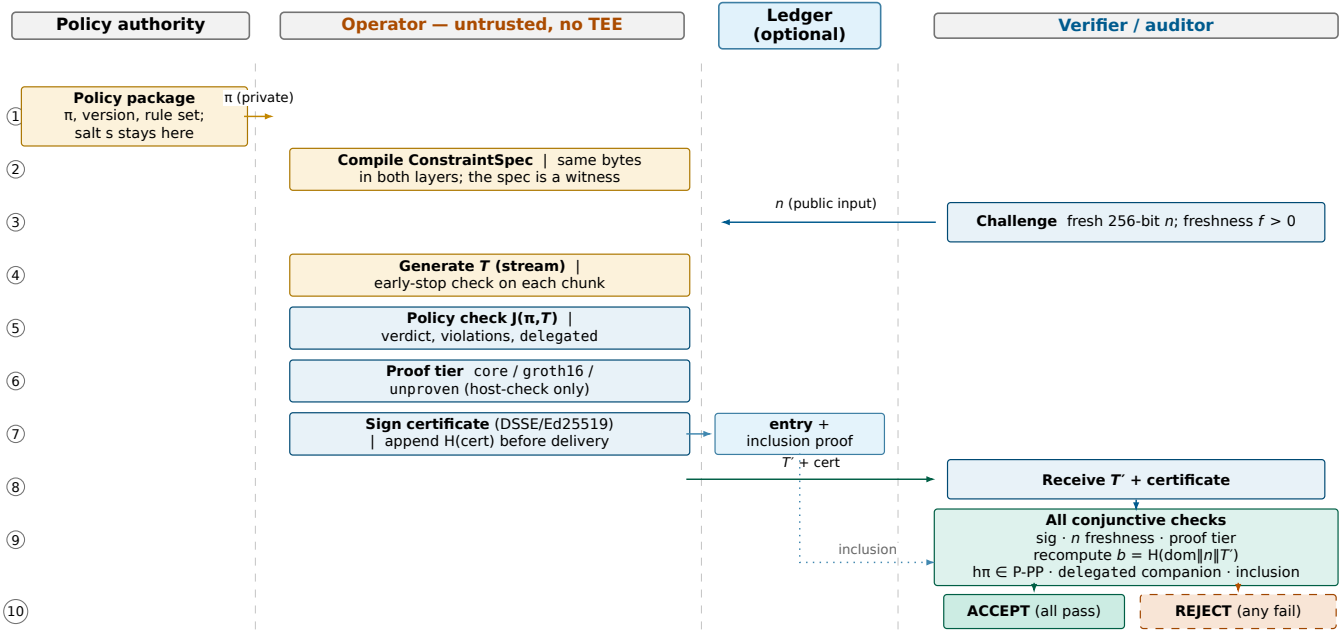


Boundaries: core is not ZK; groth16 hiding is wrapper-only, not post-quantum; unproven is host-check only; append-before-delivery is external; T'' binding remains open.